Politecnico di Torino

Cybersecurity for Embedded Systems

Trusted Platform Module (TPM) Implementation

Project Documentation

Master degree in Computer Engineering

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CHAPTER 1

Introduction

1.1 Hardware-based Security

This chapter provides an introduction to the Hardware-based Security components, which are essential to build trust at the physical level of a system; they protect against attacks that can't be mitigated by software alone.

1.1.1 Hardware Trust

Hardware trust is a critical aspect of modern computing systems, ensuring that the hardware components are genuine and have not been tampered with.

This trust is established through various mechanisms, including secure boot processes, hardware root of trust, and the use of trusted platform modules (TPMs).

1.1.2 Root of Trust

The root of trust is a foundational component in hardware security, providing a secure anchor for the system's integrity. It is responsible for verifying the authenticity of hardware and software components during the boot process. This component is needed to always behave in the expected way, because its misbehavior cannot be detected by the system itself.

1.1.3 Trust Anchor

A trust anchor serves as a fundamental element in establishing trust within a system. Rather than relying on external validation, such as a certificate authority, a trust anchor is typically embedded directly into hardware or software, or provisioned securely through dedicated channels.

1.2 Trusted Platform Module (TPM)

The Trusted Platform Module (TPM) is a specialized hardware component, specifically a microcontroller, designed to enhance the security of computing devices. It can be a discrete chip on the motherboard or integrated into the CPU.

It provides a secure environment for generating, storing and managing cryptographic keys, ensuring that sensitive operations are performed in a protected manner. It allows to perform platform integrity measurements, and is used also for secure boot enforcement.

It acts as a harware-based root of trust, providing a secure foundation for establishing a chain of trust for the entire system during the boot process.

It defines three kinds of Roots of Trust, cooperating to measure, store, and report trusted information:

- **Root of Trust for Measurement (RTM)**: Responsible for making inherently reliable integrity measurements of the system's state during the boot process.
- **Root of Trust for Storage (RTS)**: Provides secure storage for cryptographic keys and maintains an accurate summary of values of integrity digests and the sequence of digests.
- **Root of Trust for Reporting (RTR)**: Facilitates the reliable reporting of the system's state and integrity measurements to remote entities.

In the figure 1.1, the architecture of a TPM is shown, showing the interaction between the different Roots of Trust components.

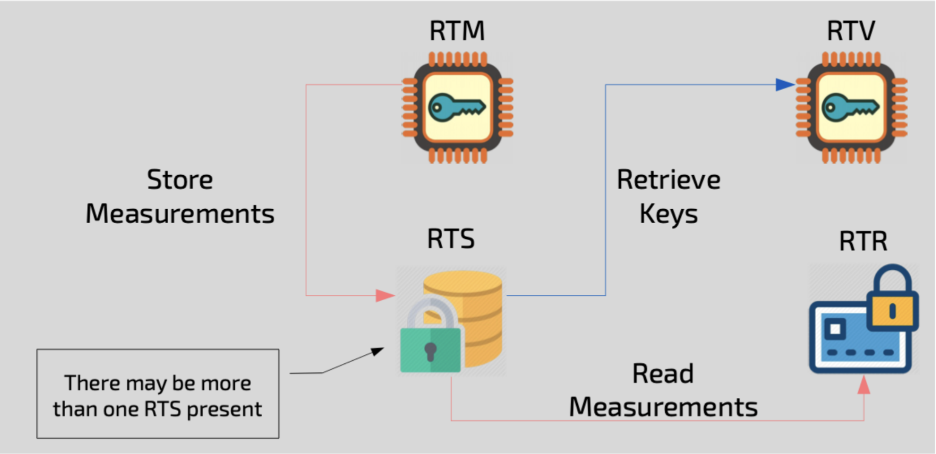

Figure 1.1: TPM Architecture and Roots of Trust

1.2.1 Certification and Attestation

Certification and attestation are essential for establishing and verifying trust in TPM-enabled platforms.

Certification uses cryptographic certificates to validate TPM components, starting with a key shipped with the TPM. This certified key forms the basis for a chain of trust, enabling secure storage and use of signing keys to prove system integrity.

Attestation allows the TPM to prove its state and authenticity to external entities, using unique identifiers (e.g., from Physical Unclonable Functions) and cryptographic evidence. Attestation covers compliance, the presence of trusted roots, and protection of key pairs.

Modern hierarchical attestation extends beyond one-time certification, enabling ongoing, layered verification of the platform's trustworthiness, including the TPM, boot process, and system configuration.

1.3 Tools for Implementation and Testing

To implement and test the TPM from the hardware side, we used QEMU.

QEMU is an open-source machine emulator and virtualizer that allows you to test and run harware platforms and operating systems in a virtualized environment. It supports a wide range of architectures and can emulate various hardware components, including CPUs, memory, and peripherals. It is widely used for development, testing, and debugging purposes, providing a flexible and efficient way to simulate hardware systems. It is particularly useful for testing firmware and software in a controlled environment before deploying it on actual hardware.

In this case it was used to emulate the hardware platform, adding the TPM component to the emulated system.

CHAPTER 2

Implementation

The implementation of the Trusted Platform Module (TPM) involves the development of both hardware and firmware components.

2.1 Hardware

2.2 Firmware

CHAPTER 3

Results

CHAPTER 4

Further Improvements

CHAPTER 5

References